

ALICE CARIBONI

Software Engineer

Interested in Firmware, Operating systems, Code Optimization

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📍 Grenoble, France

in [Alice Cariboni](#)

ABOUT ME

Hello! I am Alice, a contortionist, circus artist, and computer engineer. Though these might seem like very different worlds, I see a common denominator: solving problems, staying focused, exercising patience, and conducting long-term research that may or may not yield results depending on the approach taken. I define myself as a curious and creative person, always eager to learn. Over the next year, I'd like to develop my skills further in firmware development and operating systems.

LANGUAGES

Italian: Native speaker

English: Full professional proficiency, TOEIC Certification C1

French: Beginner

EXPERIENCE

Software Engineer – HPC R&D, *Eviden (Atos Group)*

📅 Jan 2025 – Present

- Development and optimization in C of low-level MPI components, in particular components of one-sided communications and use of performance evaluation tools (PAPI).
- Development of a collective engine component that plugs into Open MPI to abstract and select messaging layers and implementation of a shared-memory collective component for UCC utilizing shared memory for efficient inter-process data movement.
- Implementation of a code coverage pipeline for Open MPI using **Bash** scripting and OSU benchmarks.
- Management of delivery pipelines for **DOCA OFED**, including CI/CD automation, packaging, release processes, and automation of deployment and installation processes for exascale clusters using **Ansible**, **Kubernetes**, and occasional **Bash** scripting.
- Testing and validation of all code and components on real HPC clusters using **SLURM** scheduler.

Master Thesis, *Diaxxo (ETH Zurich)*

📅 Sept 2023 – March 2024

- Firmware development in C for multitasking RTOS on ESP32: concurrent tasks for real sensor acquisition (ADC thermistors), safety monitoring, serial communication, and PCR execution with deterministic timing constraints.
- Temperature control strategies combining machine learning and PID autotuning within hybrid bang-bang and feedback control architectures.
- Virtual temperature sensor module with thermal dynamics modeling for hardware-independent testing and fail-safe mechanisms.
- Python backend tools for serial communication, parameter tuning, data visualization, and firmware validation.

Software Developer, *WEART*

📅 March 2022 - September 2022

Development of UWP Applications in C#/XAML, integration of new functionalities in an existing middleware application, porting of Processing script in C#, and expansion of serial communication protocol.

EDUCATION

M.Sc. in Computer Science and Engineering, *Politecnico di Milano*

📅 Sept 2021 – April 2024

📄 Final grade: 102/110

Thesis title: “*Design and implementation of a temperature control strategy for a high-speed heating/cooling module*”.

Relevant courses: Sensor Systems, Code transformation and optimization, Internet of Things, Embedded Systems, Advanced Operating Systems.

B.Sc. in Computer Engineering, *Politecnico di Milano*

📅 Sept 2018 – Sept 2021

📄 Final grade: 96/110